

Qualification Pack



Engineer - Transmission

QP Code: PSS/Q7003

Version: 1.0

NSQF Level: 6

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PSS/Q7003: Engineer - Transmission

Brief Job Description

Engineer-Transmission is responsible for power transmission systems erection and commissioning, operation & maintenance, testing & inspection of grid substation, power transformer, transmission network, O/H Line, U/G cabling, GIS, SCADA, automation system, smart grid, metering etc. including HV and EHV systems protection.

Personal Attributes

Engineer Transmission should be alert to observe minute details at the work place. The job holder should be able to cope with difficult work environment (such as heat, noise, height etc.). He/She must also have physical ability and stamina to stand for long working hours, besides mental ability to respond to emergency situation.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1336: Working effectively with others](#)
2. [PSS/N2001: Use basic health and safety practices for power related work](#)
3. [PSS/N7003: Carry out erection & commissioning of transmission system](#)
4. [PSS/N7004: Carry out operation & maintenance of power transmission system](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Transmission
Occupation	Engineer Transmission
Country	India
NSQF Level	6
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3113



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Minimum Educational Qualification & Experience	B.E./B.Tech (Electrical)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	17/05/2017
Next Review Date	31/03/2022
NSQC Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03484
NQR Version	1.0



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PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



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- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



Qualification Pack

PSS/N2001: Use basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** identify job-site hazardous work and state possible causes of risk or accident in the workplace
- PC5.** follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),
- PC6.** follow warning signs (danger, out of service, etc.) while working with electrical systems
- PC7.** use standard safe working practices when working at heights, confined areas and trenches
- PC8.** test any electrical equipment and system using insulated testing devices before touching them
- PC9.** ensure positive isolation of electrical equipment & system as per given standards
- PC10.** recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed
- PC11.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC12.** state methods of accident prevention in the work environment of the job role
- PC13.** state location of general health and safety equipment in the workplace
- PC14.** inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder
- PC15.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa
- PC16.** inspect grid station and its equipment routinely for any signs of oil and water leakage
- PC17.** store flammable materials and machine lubricating oil safely and correctly
- PC18.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC19.** ensure proper working condition of battery and battery charger
- PC20.** maintain electrolyte level of each cell with distilled water
- PC21.** maintain proper ventilation in battery room
- PC22.** apply good housekeeping practices at all times
- PC23.** identify common hazard signs displayed in various areas



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PC24. retrieve and/or point out documents that refer to health and safety in the workplace

PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly

Fire Safety

To be competent, the user/individual on the job must be able to:

PC26. use the various appropriate fire extinguishers on different types of fires correctly

PC27. distinguish between various types of fire

PC28. demonstrate rescue techniques applied during fire hazard

PC29. demonstrate good housekeeping in order to prevent fire hazards

PC30. demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC31. demonstrate how to free a person from electrocution

PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.

PC33. demonstrate basic techniques of bandaging

PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments

PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases

PC37. demonstrate the artificial respiration and the cpr process

PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work

PC39. write accident/incident report or dictate a report to another person, and send report to person responsible

PC40. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. escalation matrix and procedures for reporting work and employment related issues

KU5. meaning of hazards and risks

KU6. health and safety hazards commonly present in the work environment and related precautions

KU7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible



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- KU8.** possible causes of risk and accident
- KU9.** methods of accident prevention
- KU10.** safe working practices when working with tools and machines
- KU11.** safe working practices while working at various hazardous sites
- KU12.** where to find all the general health and safety equipment in the workplace
- KU13.** various dangers associated with the use of electrical equipment
- KU14.** positive isolation of electrical equipment and system
- KU15.** safe handling and disposal of hazardous power plant wastes
- KU16.** use of emission and pollution control devices and measures taken to control pollution
- KU17.** various safety procedures and equipment used to work at heights, trenches and confined places
- KU18.** safe working practices specific to working with electrical equipment & system e.g. lock out/tag out, ptw, etc.
- KU19.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU20.** importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
- KU21.** precautionary activities taken to prevent fire accident
- KU22.** various causes of fire
- KU23.** techniques of using the different fire extinguishers
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire
- KU26.** emergency rescue techniques applied during a fire hazard
- KU27.** various types of safety signs and what they mean
- KU28.** appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge.
- GS2.** note down observations (if any) related to the operation/maintenance.
- GS3.** read and interpret the process required for different types of manuals for maintenance.
- GS4.** read and interpret the flowchart of all parts of an assembly.
- GS5.** read manuals and documents to understand the product-details & how they can be used.
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor.
- GS7.** effectively communicate with the team members.
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor.
- GS9.** communicate clearly with the colleague on the issues faced during query/fault.
- GS10.** follow colleague/contractor rule-based decision making process.
- GS11.** take decisions with systematic course of actions and/or response.



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- GS12.** planning and organization of tasks to meet deadlines.
- GS13.** build customer relationships and use customer centric approach.
- GS14.** seek and comprehend operation related inputs for clarification
- GS15.** find ways of modifying difficult operating stages to make it operation friendly
- GS16.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results.
- GS17.** quick approach and solution towards faults repairing.
- GS18.** critically evaluate operation parameters in relation to system normality
- GS19.** develop a holistic and comprehensive profile of grid station on segregated discrete process.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	11	46	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	-	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	-	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	-	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace	1	1	-	-
PC5. follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),	1	2	-	-
PC6. follow warning signs (danger, out of service, etc.) while working with electrical systems	1	1	-	-
PC7. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC8. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC9. ensure positive isolation of electrical equipment & system as per given standards	1	2	-	-
PC10. recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed	1	2	-	-
PC11. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	1	-	-
PC12. state methods of accident prevention in the work environment of the job role	-	2	-	-
PC13. state location of general health and safety equipment in the workplace	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder	-	2	-	-
PC15. lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa	-	2	-	-
PC16. inspect grid station and its equipment routinely for any signs of oil and water leakage	-	2	-	-
PC17. store flammable materials and machine lubricating oil safely and correctly	-	2	-	-
PC18. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	1	-	-
PC19. ensure proper working condition of battery and battery charger	1	1	-	-
PC20. maintain electrolyte level of each cell with distilled water	-	2	-	-
PC21. maintain proper ventilation in battery room	-	1	-	-
PC22. apply good housekeeping practices at all times	1	2	-	-
PC23. identify common hazard signs displayed in various areas	-	2	-	-
PC24. retrieve and/or point out documents that refer to health and safety in the workplace	-	2	-	-
PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly	-	3	-	-
Fire Safety	4	9	-	-
PC26. use the various appropriate fire extinguishers on different types of fires correctly	1	1	-	-
PC27. distinguish between various types of fire	1	2	-	-
PC28. demonstrate rescue techniques applied during fire hazard	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate good housekeeping in order to prevent fire hazards	-	2	-	-
PC30. demonstrate the correct use of a fire extinguisher	1	2	-	-
<i>Emergencies, rescue and first-aid procedures</i>	9	21	-	-
PC31. demonstrate how to free a person from electrocution	1	2	-	-
PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	-	3	-	-
PC33. demonstrate basic techniques of bandaging	1	2	-	-
PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC37. demonstrate the artificial respiration and the cpr process	1	2	-	-
PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC39. write accident/incident report or dictate a report to another person, and send report to person responsible	1	2	-	-
PC40. demonstrate correct method to move injured people and others during an emergency	1	2	-	-
NOS Total	24	76	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2001
NOS Name	Use basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022



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PSS/N7003: Carry out erection & commissioning of transmission system

Description

Engineer-Transmission is responsible for transmission systems erection and commissioning. In addition of the above material management, resource mobilization and co-ordination with seniors shall also be part of the job.

Scope

This unit/task covers the following:

Elements and Performance Criteria

Installation of power transmission network

To be competent, the user/individual on the job must be able to:

- PC1.** describe power transmission system including types of transmission system
- PC2.** differentiate various types of transmission system voltage level wise/technology (hvdc, hvac)
- PC3.** oversee the civil foundation work
- PC4.** draw and plan schematic diagrams, engineering drawings, and single line designs, diagrams, lay out plans etc.
- PC5.** plan, monitor and control installation of power transmission systems
- PC6.** review erection of ehv cable-oil filled and gas filled cable
- PC7.** monitor ehv cable joining process
- PC8.** manage installation of communication system such as plcc and opgw
- PC9.** prepare estimates and bill of quantities
- PC10.** handle the procurement function (modes and types- turnkey, supply & services etc.); material planning and handling; store handling methods
- PC11.** supervise out route survey for o/h line or u/g lines and row
- PC12.** supervise gis mapping of transmission assets
- PC13.** supervise erection and commissioning of protection system and earthing for transmission network
- PC14.** prepare mis report

Installation of grid sub-station

To be competent, the user/individual on the job must be able to:

- PC15.** plan, monitor and control erection and commissioning of air insulated substation and gas insulated substation (gis)
- PC16.** oversee erection of power transformer, switchgear, lighting arrester, ct, pt, cvt etc.
- PC17.** perform activities related to bus bar and bus coupler
- PC18.** oversee prepare transformer foundation
- PC19.** oversee installation of substation automation system (sas)
- PC20.** conduct type testing and regular testing of various equipment and relevant specifications for conducting them



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- PC21.** testing or inspection of transformer, switchgear and other equipment for pre and post commissioning
- PC22.** oversee implementation of scada
- PC23.** perform proper earthing system for substation equipment
- PC24.** ensure erection of control panels in control room

Protection system

To be competent, the user/individual on the job must be able to:

- PC25.** supervise line protection system
- PC26.** ensure substation equipment protection system
- PC27.** ensure installation of earthing system
- PC28.** setup the types of relay and relay setting
- PC29.** supervise installation of various types of breaker
- PC30.** oversee implementation of das and associated communication system
- PC31.** oversee installation of protection devices- surge protection device, over voltage protection etc.
- PC32.** coordinate and manage logistic related issues

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the organization relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** own job role, responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** how to engage with specialists for support in order to resolve incidents and service requests
- KU6.** importance of working in clean and safe environment practices and procedures
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** network layout, schematic and design drawing
- KU10.** technical specification of transmission system equipment
- KU11.** overview of policy and regulatory regime in the sector
- KU12.** design of transmission network based on prevailing planning and policy guidelines
- KU13.** overview of load flow studies, sag and tension calculation, transient studies, vibration analysis, wind pressure analysis, short circuit studies etc.
- KU14.** various type of equipment protection system
- KU15.** ratings and specifications of line, transformer, cables, fuses, switches and wires
- KU16.** handling of all machineries, equipment & vehicles
- KU17.** use of appropriate judgment and initiative pertaining to work methods and tools



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- KU18.** technical manuals, blueprints, schematics, diagrams, plans, specifications estimate time, material and equipment needed to complete assignments
- KU19.** standard procedures, how to deal with electric shocks and electrocutions to rescue and minimize damage and harm
- KU20.** quality parameters, quality assessment based on physical parameters

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information related to work
- GS2.** note down observations (if any) related to the process
- GS3.** use IT
- GS4.** prepare estimates
- GS5.** document and keep records
- GS6.** read and interpret the process required for various types of operations
- GS7.** read and interpret and process flowchart for all operations
- GS8.** read manuals and operation documents to understand the Equipment used into operation
- GS9.** discuss task lists, schedules and activities with the seniors
- GS10.** effectively communicate with the team members
- GS11.** communicate clearly with the customer
- GS12.** follow organization rule-based decision making process
- GS13.** take decision with systematic course of actions and/or response
- GS14.** lead
- GS15.** understand importance of proper documentation
- GS16.** planning and organization of tasks to meet deadlines
- GS17.** build customer relationships and use customer centric approach
- GS18.** seek and comprehend operation related inputs for clarification
- GS19.** find ways of modifying difficult operating stages to make them operation friendly
- GS20.** apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS21.** critically evaluate operation parameters in relation to product features intended
- GS22.** develop holistic and comprehensive profile of products based on segregated discrete process stages

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation of power transmission network</i>	13	35	-	-
PC1. describe power transmission system including types of transmission system	1	2	-	-
PC2. differentiate various types of transmission system voltage level wise/technology (hvdc, hvac)	1	1	-	-
PC3. oversee the civil foundation work	1	2	-	-
PC4. draw and plan schematic diagrams, engineering drawings, and single line designs, diagrams, lay out plans etc.	1	3	-	-
PC5. plan, monitor and control installation of power transmission systems	1	2	-	-
PC6. review erection of ehv cable-oil filled and gas filled cable	1	3	-	-
PC7. monitor ehv cable joining process	1	3	-	-
PC8. manage installation of communication system such as plcc and opgw	-	4	-	-
PC9. prepare estimates and bill of quantities	1	3	-	-
PC10. handle the procurement function (modes and types- turnkey, supply & services etc.); material planning and handling; store handling methods	1	3	-	-
PC11. supervise out route survey for o/h line or u/g lines and row	1	3	-	-
PC12. supervise gis mapping of transmission assets	1	2	-	-
PC13. supervise erection and commissioning of protection system and earthing for transmission network	1	2	-	-
PC14. prepare mis report	1	2	-	-
<i>Installation of grid sub-station</i>	9	20	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. plan, monitor and control erection and commissioning of of air insulated substation and gas insulated substation (gis)	1	2	-	-
PC16. oversee erection of power transformer, switchgear, lighting arrester, ct, pt, cvt etc.	1	2	-	-
PC17. perform activities related to bus bar and bus coupler	1	1	-	-
PC18. oversee prepare transformer foundation	1	1	-	-
PC19. oversee installation of substation automation system (sas)	1	3	-	-
PC20. conduct type testing and regular testing of various equipment and relevant specifications for conducting them	1	3	-	-
PC21. testing or inspection of transformer, switchgear and other equipment for pre and post commissioning	1	3	-	-
PC22. oversee implementation of scada	1	2	-	-
PC23. perform proper earthing system for substation equipment	1	1	-	-
PC24. ensure erection of control panels in control room	-	2	-	-
<i>Protection system</i>	8	15	-	-
PC25. supervise line protection system	1	1	-	-
PC26. ensure substation equipment protection system	1	2	-	-
PC27. ensure installation of earthing system	1	1	-	-
PC28. setup the types of relay and relay setting	1	3	-	-
PC29. supervise installation of various types of breaker	1	1	-	-
PC30. oversee implementation of das and associated communication system	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC31. oversee installation of protection devices-surge protection device, over voltage protection etc.	1	2	-	-
PC32. coordinate and manage logistic related issues	1	3	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N7003
NOS Name	Carry out erection & commissioning of transmission system
Sector	Power
Sub-Sector	Transmission
Occupation	Transmission System Erection, Commissioning, Operation & Maintenance
NSQF Level	6
Credits	TBD
Version	1.0
Last Reviewed Date	17/05/2017
Next Review Date	31/03/2022
NSQC Clearance Date	NA



Qualification Pack

PSS/N7004: Carry out operation & maintenance of power transmission system

Description

This section covers the operation and maintenance of transmission systems like transmission line, grid substation, power transformer, EHV U/G cable, switchgear and associated components etc. ensuring healthy operation of transmission system equipment.

Scope

This unit/task covers the following:

- operation & maintenance
- Transmission
- testing of the system
- repairing & replacement

Elements and Performance Criteria

Operation & maintenance - Transmission

To be competent, the user/individual on the job must be able to:

- PC9.** undertake operation of grid substation and transmission lines
- PC10.** check all the intersections & joints (termination) in the line or cable
- PC11.** locate the conduit, cables & other underground system to perform maintenance work
- PC12.** monitor performance of critical system such as sas, rtu, remote monitoring unit (rmu) and other automation system
- PC13.** supervise and carry out preventive, breakdown and routine maintenance for lines and substation equipment
- PC14.** monitor problem and keep the superiors informed about progress or any delays in resolving the problem
- PC15.** prepare estimates, bill of quantity for carrying out maintenance activity
- PC16.** do asset management of power transmission system
- PC17.** conduct load management, grid stability, frequency, load dispatch, feeder loading etc. national occupational
- PC18.** co-ordinate resources, mobilize teams, build teams, resolve interpersonal issues, manage logistics
- PC19.** create sops, schedules, maintenance schedules
- PC1.** inspect power transformer / switchgear/transmission lines for preventive and breakdown maintenance
- PC2.** routinely patrol transmission line, identifying hot spot and maintain required clearance
- PC3.** supervise and inspect maintenance of ehv u/g cable
- PC4.** undertake fault detection
- PC5.** do relay and breaker maintenance



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- PC6.**
- monitor running parameter of critical equipment
 - T/F, S/G and other electrical equipment

PC7. supervise hotline and cold line maintenance

PC8. supervise hotline washing of insulator

Testing of the system

To be competent, the user/individual on the job must be able to:

PC20. test the system parameter to know abnormal condition of the system

PC21. test the healthiness of connected equipment

Repairing & replacement

To be competent, the user/individual on the job must be able to:

PC22. supervise and carry out repair and replacement of faulty/ unhealthy equipment

PC23. troubleshoot a faulty system

PC24. do r&m of the existing unhealthy equipment/system

PC25. supervise carry out general routine repair work

PC26. implement technical change in equipments/systems

PC27. co-ordinate resources, mobilize teams, build teams, resolve interpersonal issues, manage logistics

PC28. create sops, schedules, maintenance schedules

PC26. implement technical change in equipment's/systems

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the organization relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** how to engage with specialists for support in order to resolve incidents and service requests
- KU6.** importance of working in clean and safe environment practices and procedures
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** tools and technologies available for conducting maintenance activities
- KU10.** load management, grid stability, frequency, load dispatch, etc.
- KU11.** relay and breaker maintenance
- KU12.** hotline and cold line maintenance of transmission system
- KU13.** emergency restoration procedures
- KU14.** technical manuals, blueprints, schematics, diagrams, plans, specifications estimate time, material and equipment needed to complete assignments



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- KU15.** common electricity terminology and correct interpretation of the same terminology: e.g. current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
- KU16.** ratings and specifications of cables, fuses, switches and wires
- KU17.** machineries, equipment & vehicles
- KU18.** use of appropriate judgment and initiative pertaining to work methods and tools
- KU19.** asset management
- KU20.** disaster management

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the supervisor or engineer
- GS2.** note down observations (if any) related to the process of installation, operation and maintenance of power transmission system
- GS3.** read and interpret the process required for various types of operations related to power transmission system
- GS4.** read and interpret and process flowchart for all operations related to power transmission system
- GS5.** read manuals and operation documents to understand the equipment used into operation
- GS6.** discuss task lists, schedules and activities with the supervisor
- GS7.** effectively communicate with the team members
- GS8.** communicate clearly with the customer on the issues faced during query/fault
- GS9.** follow organization rule-based decision making process
- GS10.** take decision with systematic course of actions and/or response
- GS11.** planning and organization of tasks to meet deadlines
- GS12.** build customer relationships and use customer centric approach
- GS13.** seek and comprehend operation related inputs for clarification
- GS14.** find ways of modifying difficult operating stages to make it operation friendly
- GS15.** apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS16.** lead and manage
- GS17.** critically evaluate operation parameters in relation to equipment features
- GS18.** develop holistic and comprehensive profile of products based on segregated discrete process stages



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operation & maintenance - Transmission</i>	19	47	-	-
PC9. undertake operation of grid substation and transmission lines	-	3	-	-
PC10. check all the intersections & joints (termination) in the line or cable	-	2	-	-
PC11. locate the conduit, cables & other underground system to perform maintenance work	1	2	-	-
PC12. monitor performance of critical system such as sas, rtu, remote monitoring unit (rmu) and other automation system	1	2	-	-
PC13. supervise and carry out preventive, breakdown and routine maintenance for lines and substation equipment	1	4	-	-
PC14. monitor problem and keep the superiors informed about progress or any delays in resolving the problem	1	2	-	-
PC15. prepare estimates, bill of quantity for carrying out maintenance activity	2	3	-	-
PC16. do asset management of power transmission system	2	2	-	-
PC17. conduct load management, grid stability, frequency, load dispatch, feeder loading etc. national occupational	1	3	-	-
PC18. co-ordinate resources, mobilize teams, build teams, resolve interpersonal issues, manage logistics	1	2	-	-
PC19. create sops, schedules, maintenance schedules	1	2	-	-
PC1. inspect power transformer / switchgear/transmission lines for preventive and breakdown maintenance	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC2. routinely patrol transmission line, identifying hot spot and maintain required clearance	1	3	-	-
PC3. supervise and inspect maintenance of ehv u/g cable	1	3	-	-
PC4. undertake fault detection	1	3	-	-
PC5. do relay and breaker maintenance	1	2	-	-
PC6. • monitor running parameter of critical equipment • T/F, S/G and other electrical equipment	1	1	-	-
PC7. supervise hotline and cold line maintenance	1	3	-	-
PC8. supervise hotline washing of insulator	1	2	-	-
<i>Testing of the system</i>	2	4	-	-
PC20. test the system parameter to know abnormal condition of the system	1	2	-	-
PC21. test the healthiness of connected equipment	1	2	-	-
<i>Repairing & replacement</i>	9	19	-	-
PC22. supervise and carry out repair and replacement of faulty/ unhealthy equipment	1	2	-	-
PC23. troubleshoot a faulty system	2	3	-	-
PC24. do r&m of the existing unhealthy equipment/system	1	3	-	-
PC25. supervise carry out general routine repair work	1	3	-	-
PC26. implement technical change in equipments/systems	1	3	-	-
PC27. co-ordinate resources, mobilize teams, build teams, resolve interpersonal issues, manage logistics	1	3	-	-
PC28. create sops, schedules, maintenance schedules	2	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. implement technical change in equipment's/systems	1	3	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N7004
NOS Name	Carry out operation & maintenance of power transmission system
Sector	Power
Sub-Sector	Transmission
Occupation	Transmission System Erection, Commissioning, Operation & Maintenance
NSQF Level	6
Credits	TBD
Version	1.0
Last Reviewed Date	17/05/2017
Next Review Date	31/03/2022
NSQC Clearance Date	NA

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.



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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2001.Use basic health and safety practices for power related work	24	76	-	-	100	15
PSS/N7003.Carry out erection & commissioning of transmission system	30	70	-	-	100	40
PSS/N7004.Carry out operation & maintenance of power transmission system	30	70	-	-	100	40
Total	114	286	-	-	400	100



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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
KW	Kilowatt



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.